

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Technology (B Tech)

Semester-II University Examination May 2010

B.Tech. (CL/EC/ME/ CE/EE/IT)

MA 102 Engineering Mathematics-II

Date: 10.05.10, Monday

Time: 12:30 p. m to 03:30 p. m.

Maximum Marks: 70

Instructions:

1. Attempt all the questions in each section.
2. Assume suitable additional data if required.
3. Figures to the right side indicate the full marks.

SECTION-I

Q-1 (a) Solve: $\left[y(1+x^{-1}) + \sin y \right] dx + (x + \log x + x \cos y) dy = 0$ [04]

(b) Solve $y'' - 2y' + y = \frac{e^x}{x^4}$ by method of variation of parameters. [04]

(c) Solve: $3p^2 - 2q^2 = 4pq$ [03]

Q-2 (i) Solve: $(x+1)^2 \frac{d^2 y}{dx^2} - 3(x+1) \frac{dy}{dx} + 4y = x^2$ [04]

(ii) Solve: $y'' + y' - 2y = e^{-t} \sin t$ [04]

(iii) Solve the initial-value problem: $x^2 y' + xy = 1$, $x > 0$, $y(1) = 2$. [04]

OR

Q-2 (i) Solve: $x^3 \frac{d^3 y}{dx^3} + 2x^2 \frac{d^2 y}{dx^2} + 2y = 10 \left(x + \frac{1}{x} \right)$ [04]

(ii) If the population of a country doubles in 50 years, in how many years will it treble, assuming that the rate of increase is proportional to the number of inhabitants? [04]

(iii) Solve: $\frac{dy}{dx} + \frac{2}{x} y = -x^2 y^2 \cos x$ [04]

Q-3 (a) Solve the simultaneous linear equations: $\frac{dx}{dt} - 7x + y = 0$, $\frac{dy}{dt} - 2x - 5y = 0$ [04]

(b) Solve: $z^2(p^2 + q^2 + 1) = c^2$, where c is a constant [04]

(c) Solve: $dx + 3dy = 5z + \tan(y - 3x)$ [04]

OR

Q-3 (a) Solve: $z(x+y)dx + z(x-y)dy = x^2 + y^2$ [04]

(b) Solve: $p^2 + q^2 = z^2(x+y)$ [04]

(c) Solve the simultaneous linear equations: $\frac{dx}{dt} + \omega y = 0$, $\frac{dy}{dt} - \omega x = 0$ [04]

SECTION-II

Q-4 (a) Assuming standard notations, Prove that $\operatorname{erf}(x) + \operatorname{erfc}(x) = 1$

[03]

(b) By changing the order of integration, evaluate $\int_0^1 \int_x^1 \sin(\pi y^2) dy dx$

[03]

(c) Verify that the matrix $A = \begin{bmatrix} 0 & 1 & 2 \\ 2 & -3 & 0 \\ 1 & 1 & -1 \end{bmatrix}$ satisfies the Cayley-Hamilton theorem and

[05]

hence obtain A^{-1} .

Q-5 (i) Prove that $\int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} \times \int_0^{\pi/2} \sqrt{\sin \theta} d\theta = \pi$

[03]

(ii) If $A = \begin{bmatrix} -1 & 2+i & 5-3i \\ 2-i & 7 & 5i \\ 5+3i & -5i & 2 \end{bmatrix}$, Show that (i) A^2 is Hermitian (ii) iA is Skew-Hermitian.

[05]

(iii) Find the Volume of the region bounded by the surfaces $x^2 + y^2 + z^2 = 1$ & $x^2 + y^2 = \frac{1}{4}$.

[04]

OR

Q-5 (i) Prove that $\int_0^\infty \frac{t^2}{1+t^4} dt = \frac{\pi}{\sqrt{2}}$

[03]

(ii) Define : (i) Hermitian matrix (ii) Unitary Matrix.

[05]

Show that the Matrix B^*AB is Hermitian, where $A = \begin{pmatrix} 3 & 2-i \\ 2+i & 4 \end{pmatrix}$ & $B = \begin{pmatrix} 1 & 1+i \\ 1-i & 0 \end{pmatrix}$

(iii) Evaluate: $\iiint_V \sqrt{3x^2 + 3z^2} dV$, where V is the solid bounded by $y = 2x^2 + 2z^2$ and the plane $y = 8$.

[04]

Q-6 (a) For Legendre Polynomials $P_m(x), P_n(x)$, show that $\int_{-1}^1 P_m(x) P_n(x) dx = 0$, if $m \neq n$.

[04]

(b) Assuming standard notations, Prove that $\frac{d}{dx} [x J_n(x) J_{n+1}(x)] = x [J_n^2(x) - J_{n+1}^2(x)]$

[04]

(c) Evaluate: $\iint_D (x-y) dA$, where D is the triangle with vertices $(0,0), (1,0)$ and $(2,1)$.

[04]

OR

Q-6 (a) For integer n , show that $\int_{-1}^1 P_n^2(x) dx = \frac{2}{2n+1}$, where $P_n(x)$ is a Legendre Polynomial.

[04]

(b) Assuming standard notations, Prove : $2^2 J_n''(x) = J_{n-2}(x) - 2J_n(x) + J_{n+2}(x)$.

[04]

(c) Evaluate: $\iiint_V 16z dV$, where V is the upper half of the sphere $x^2 + y^2 + z^2 = 1$.

[04]